Chapter 30

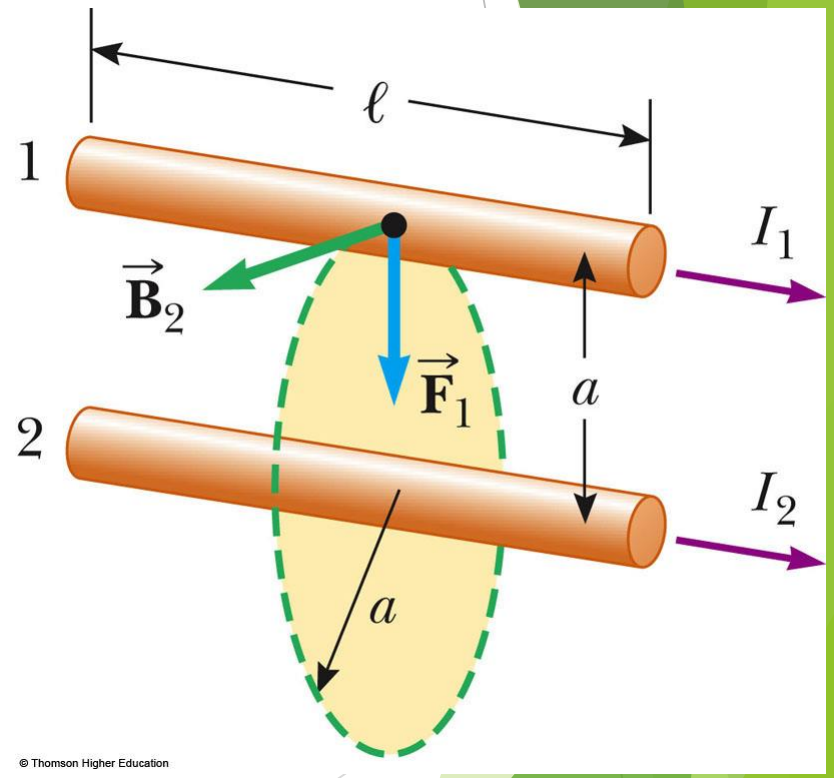
Sources of the Magnetic Field

Sec 30.2: The Magnetic Force Between Two Parallel Conductors

- Two parallel wires each carry a steady current
- The force between the two wires measured by:

$$F_1 = \frac{\mu_o I_1 I_2}{2\pi a} \ell$$

- The constant mo is called the **permeability of free space**
- $\mu_o = 4\pi \times 10^{-7}$ T· m / A



Magnetic Force Between Two Parallel Conductors, cont.

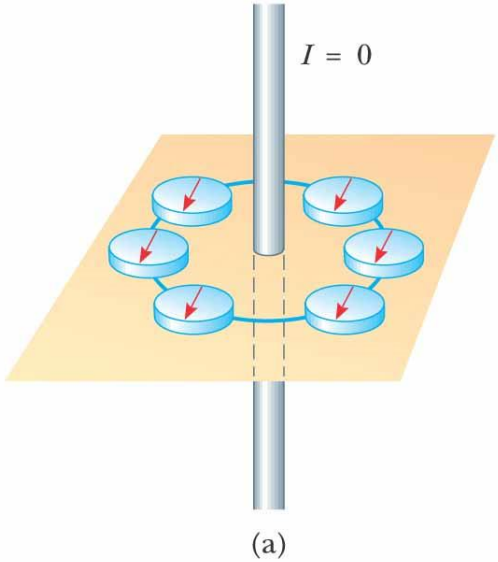
- Parallel conductors carrying currents in the same direction attract each other
- Parallel conductors carrying current in opposite directions repel each other

- The result is often expressed as the magnetic force between the two wires, F_B
- This can also be given as the force per unit length:

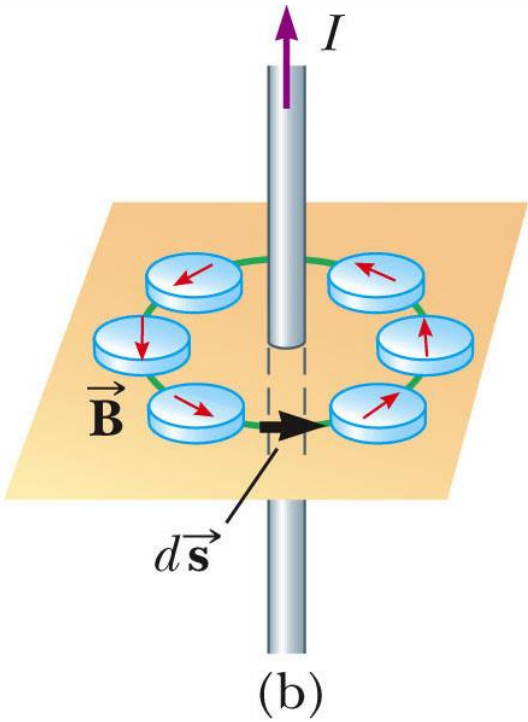
$$\frac{F_B}{\ell} = \frac{\mu_o I_1 I_2}{2\pi a}$$

Sec 30.3: Ampère's Law

- **<u>Magnetic Field of a Wire:</u>**
- A compass can be used to detect the magnetic field
- When there is no current in the wire, there is no field due to the current
- The compass needles all point toward the Earth's north pole
  - Due to the Earth's magnetic field



- Here the wire carries a strong current
- The compass needles deflect in a direction tangent to the circle
- This shows the direction of the magnetic field produced by the wire



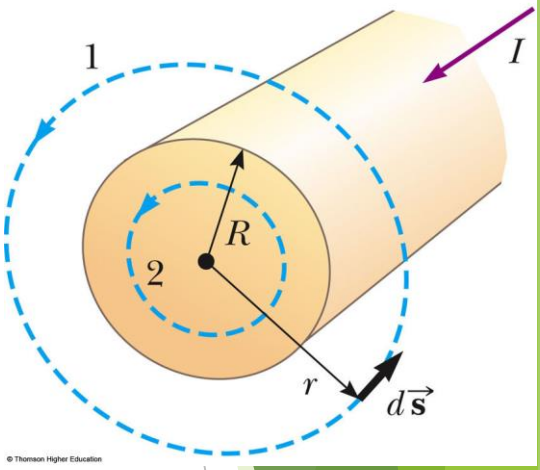
Ampere's Law

- The product of $\vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$ can be evaluated for small length elements $d\vec{\mathbf{s}}$ on the circular path defined by the compass needles for the long straight wire
- Ampere's law states that the line integral of $\vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$ around any closed path equals $\mu_o I$ where I is the total steady current passing through any surface bounded by the closed path:

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \mu_o I$$

- Ampere's law describes the creation of magnetic fields by all continuous current configurations
- Put the thumb of your right hand in the direction of the current through the amperian loop and your fingers curl in the direction you should integrate around the loop

Field Due to a Long Straight Wire



- Want to calculate the magnetic field at a distance *r* from the center of a wire carrying a steady current I
- The current is uniformly distributed through the cross-section of the wire
- Outside of the wire, $r > R$

$$B = \frac{\mu_o I}{2\pi r}$$

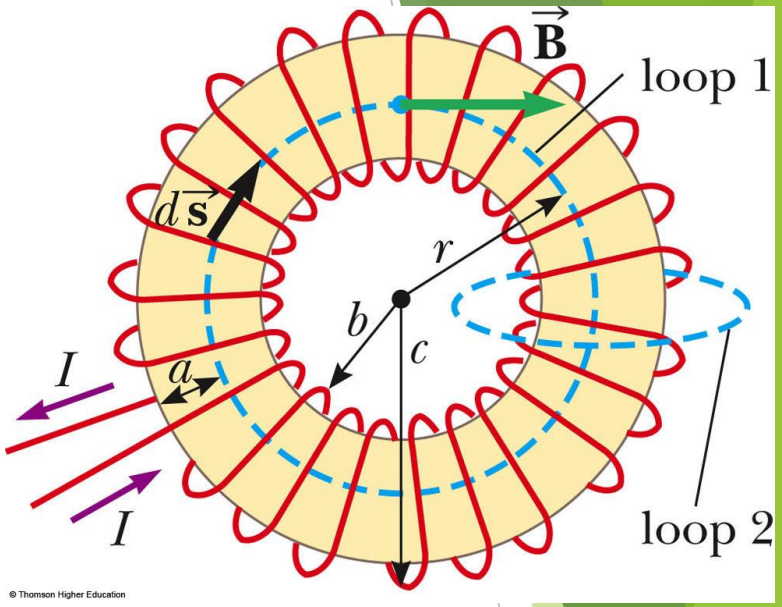
- Inside the wire, $r < R$

$$B = \left(\frac{\mu_o I}{2\pi R^2}\right) r$$

Magnetic Field of a Toroid

- Find the field at a point at distance *r* from the center of the toroid
- The toroid has *N* turns of wire

$$B = \frac{\mu_o N I}{2\pi r}$$

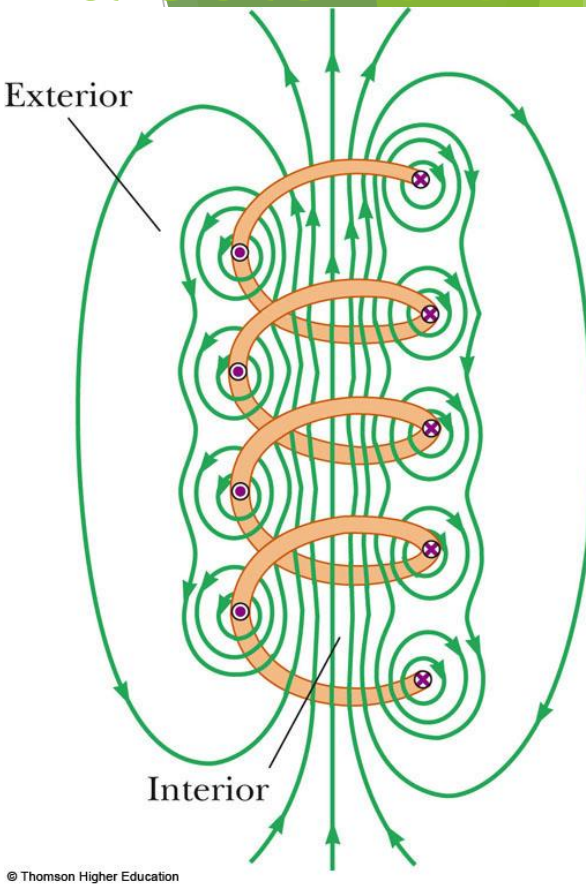


Sec 30.4: The Magnetic Field of a Solenoid

- A **solenoid** is a long wire wound in the form of a helix
- A reasonably uniform magnetic field can be produced in the space surrounded by the turns of the wire
  - The ***interior*** of the solenoid
- Solving Ampere's law for the magnetic field is

$$B = \mu_o \frac{N}{\ell} I = \mu_o n I$$

  - n = N / ℓ is the number of turns per unit length
- This is valid only at points near the center of a very long solenoid

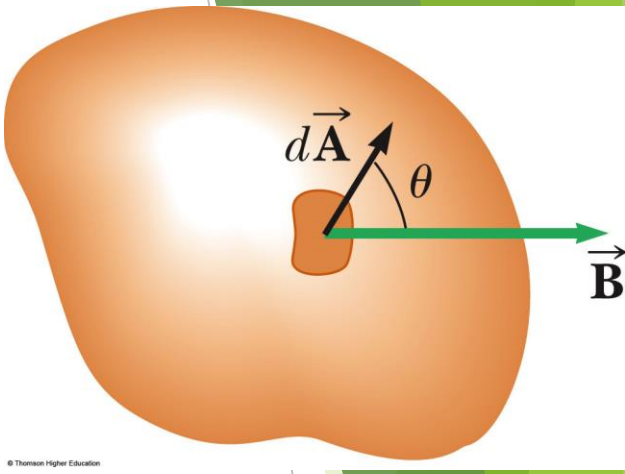


Sec 30.5: Gauss's Law in Magnetism

- The magnetic flux associated with a magnetic field is defined in a way similar to electric flux
- Consider an area element *dA* on an arbitrarily shaped surface
- The magnetic field in this element is $\vec{\mathbf{B}}$
- $d\vec{\mathbf{A}}$ is a vector that is perpendicular to the surface
- $d\vec{\mathbf{A}}$ has a magnitude equal to the area dA
- The magnetic flux F_B is

$$\Phi_B = \int \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}}$$

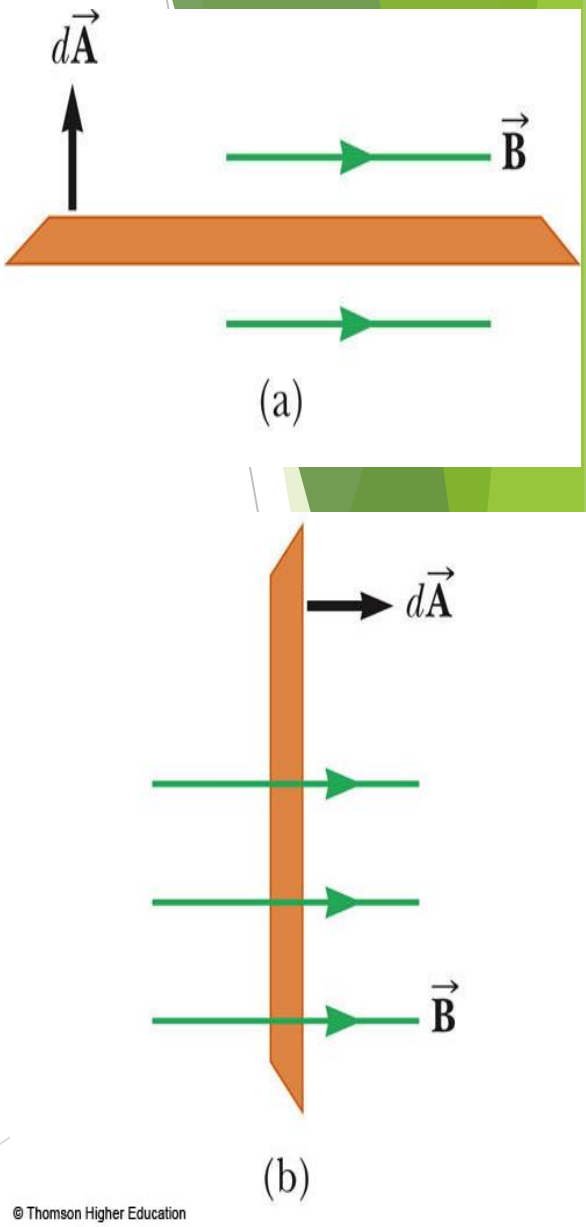
- The unit of magnetic flux is T.m^2 = Wb
  - Wb is a weber



Magnetic Flux Through a Plane

- A special case is when a plane of area *A* makes an angle θ with $d\vec{\mathbf{A}}$
- The magnetic flux is $\Phi_B = BA\cos\theta$
- In this case, the field is parallel to the plane, and $\Phi = 0$
- In this case, the field is perpendicular to the plane, and $\Phi = BA$
- This will be the maximum value of the flux
- Magnetic fields do not begin or end at any point
  - The number of lines entering a surface equals the number of lines leaving the surface
- **Gauss' law in magnetism** says the magnetic flux through any closed surface is always zero:

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0$$



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